RESEARCH BRIEFING

Technical identifiers in the OBIS Parquet lake

Which columns to mint, how they relate to `(dataset_id, DwC ID)` joins, and how GBIF does it

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Context: `separate-tables` prototype; GBIF `gbifID` / DwC-A `id`/`coreid`; gbif/dwc-dp#159

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1 Executive summary

Mint technical identifiers into Parquet **now** if multi-dataset DwC-A / DwC-DP downloads are a product goal — the same reason GBIF mints `gbifID` and remints package structural keys (159). Keep publisher Darwin Core IDs untouched. Composite `(dataset_id, occurrenceID|eventID)` joins remain valid for DuckDB analytics; they are not enough inside one zip that concatenates many datasets.

> **Bottom line**
>
> Add `event_pk` / `occurrence_pk` (and matching FKs on extension rows) to every lake table at land time. Keep `dataset_id` and DwC ID fields as data. Use the `*_pk` columns for multi-dataset packaging and, optionally, for DuckDB joins. Map `occurrence_pk` → search `obisID` (GBIF's `gbifID` role).

2 Why not composites alone

The prototype joins on `(dataset_id, DwC ID)` inside per-dataset folders. That works for DuckDB subsets. It fails for **one** archive spanning many datasets:

- `occurrenceID = 123` in two datasets → colliding package keys.
- Extension `coreid = <occurrenceID>` is not unique after concatenation.
- Frictionless / DwC-A validators expect structural PKs, not an informal "also filter by dataset_id" rule.

GBIF already separates these roles: `gbifID` for search; `id`/`coreid` (DwC-A) or `_pk`/`_fk` (DwC-DP #159) for package joins; `occurrenceID` left as publisher data.

3 Parquet fields to add

Mint at land / interpret time. Values are aggregator-owned integers or ULIDs, unique within the lake (or reminted unique within each download zip). Never overwrite DwC fields.

3.1 Every table (envelope)

Field	Required	Role
`dataset_id`	yes	Existing. Dataset UUID; provenance and lake partitioning.
`source / verbatim`	yes	Existing. Publisher strings — including DwC IDs.
`interpreted`	yes	Existing. Typed view — including DwC IDs for QC/filters.

Table 1: Columns on all Parquet row types.

3.2 `event.parquet`

Field	Required	Role
`event_pk`	yes	**Minted.** Primary structural key for this event row.
`parent_event_fk`	no	**Minted.** Points at parent row's `event_pk` when `parentEventID` resolves inside the dataset; null at roots / unresolved.
`interpreted.eventID`	yes*	Publisher semantic ID (*required by DwC when present in source). Not the join key for multi-dataset zips.
`interpreted.parentEventID`	no	Publisher parent pointer; keep for hierarchy display and composite walks.

Table 2: Technical IDs on events.

3.3 `occurrence.parquet`

Field	Required	Role
`occurrence_pk`	yes	**Minted.** Primary structural key; also the natural value to expose as API `obisID` / ES document id (1 :1 or mapped).
`event_fk`	no*	**Minted.** Points at `event.event_pk` when the occurrence is linked to an event (*required for Event-core datasets).
`interpreted.occurrenceID`	yes*	Publisher semantic ID.
`interpreted.eventID`	no	Publisher event link; keep alongside `event_fk`.

Table 3: Technical IDs on occurrences.

3.4 Extension tables

Apply the same pattern to every extension file (`event_measurementorfact`, `occurrence_measurementorfact`, `event_dnaderiveddata`, `occurrence_dnaderiveddata`, multimedia, …):

Field	Required	Role
`row_pk`	yes	**Minted.** Unique key for this extension row (needed when packing DwC-DP or citing a MoF row).
`event_fk`	if event-level	**Minted.** → `event.event_pk`.
`occurrence_fk`	if occurrence-level	**Minted.** → `occurrence.occurrence_pk`.
DwC / extension ID fields	as published	e.g. `measurementID` if present — data only, not the structural join.

Table 4: Technical IDs on extension rows.

Split event- vs occurrence-level files as today; the FK tells you which parent table to join, so you do not rely on densified copies.

3.5 Optional later (DwC-DP-shaped lake)

When landers grow Agent / Survey / Material tables, add the matching pair per table: `agent_pk`, `survey_pk`, `material_pk`, plus FKs on rows that point at them (`recorded_by_fk`, …). Same #159 rule: remintable structural columns; semantic `agentID` etc. stay in `interpreted`.

4 How joins work with these fields

Consumer	Join
DuckDB subset (prototype)	Still fine: `(dataset_id, eventID|occurrenceID)`. Or switch to `event_fk = event_pk` once minted.
Event hierarchy walk	`occurrence.event_fk` → `event.event_pk`; recurse `parent_event_fk`.
Extension attach	`mof.occurrence_fk` → `occurrence.occurrence_pk` or `mof.event_fk` → `event.event_pk`.
Multi-dataset DwC-A zip	Map `occurrence_pk` → file `id`; extension `occurrence_fk` / `event_fk` → `coreid`. Remint per zip if you want zip-local integers.
Multi-dataset DwC-DP zip	Copy (or remint) `*_pk` / `*_fk` into package columns; leave DwC ID terms unchanged (#159).
Elasticsearch / API	`obisID` = `occurrence_pk` (or a stable mapping table if you remint package keys separately).

Table 5: Join recipe by consumer.

5 Comparison with GBIF

OBIS Parquet	GBIF	Notes
`occurrence_pk` / `obisID`	`gbifID`	Search + citation; remappable on major ID churn.
`event_pk`	event structural id in pipelines / DwC-DP `event_pk`	Needed for event index + Event-core packaging.
`event_fk` / `occurrence_fk` / `parent_event_fk`	`coreid` / `_fk`	Package and lake joins.
`dataset_id`	`datasetKey`	Already present.
`interpreted.occurrenceID`	`occurrenceID`	Publisher data — never overwrite.

Table 6: OBIS Parquet column ↔ GBIF analogue.

6 Policy

1. **Mint into the lake at land time** — do not wait for the packager if multi-dataset downloads are in scope.
2. **Never put technical values into DwC ID fields.**
3. **QC** duplicates of `(dataset_id, occurrenceID)` and unresolved `parentEventID` / `eventID` links at land time; structural FKs make those failures visible.
4. **Packager may remint** `_pk`/`_fk` again inside a download zip (#159); lake keys are the default source, not a sacred global IRI.

7 Short answers

1. **Need technical IDs?** Yes — for multi-dataset DwC-A/DwC-DP and for stable `obisID`.
2. **Drop composite joins?** No — keep them for provenance/QC; prefer `*_pk`/`*_fk` for packaging.
3. **Which Parquet fields?** `event_pk`, `parent_event_fk`, `occurrence_pk`, `event_fk` on occurrences, and on each extension row `row_pk` plus `event_fk` or `occurrence_fk`.

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